

Sovannratana Khek

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SUMMARY

Computer science graduate student (B.S./M.S., Dec 2026) with full-stack development experience and strong fundamentals in backend engineering, cloud infrastructure, and systems programming. Proven ability to design and ship RESTful APIs, integrate third-party services, and deliver functional products under deadline pressure. Seeking a new grad software engineering role.

EDUCATION

Illinois Institute of Technology — Chicago, IL — *Expected: December 2026*

Combined B.S. & M.S. in Computer Science | GPA: 3.2

Coursework: Data Structures & Algorithms, Database Organization, Operating Systems, Cryptography, Data Communications, Software Engineering, OOP, Parallel Computing

EXPERIENCE

Computer Science Lead | Exelon Summer Institute / Illinois Institute of Technology | Chicago, IL — *Summers 2023–2025*

- Designed and delivered a hands-on CS curriculum for a multi-week summer program, integrating programming, problem-solving, and applied projects for 20+ students per cohort.
- Collaborated with a cross-disciplinary team to align CS content with other subject areas, improving curriculum cohesion as measured by student surveys.
- Led interactive lab sessions and project-based activities, increasing student engagement tracked via class participation logs.
- Managed course materials and student participation, coordinating with TAs to ensure consistent instruction and timely feedback.

PROJECTS

Oracle Lens — AI Coaching App (Devpost Hackathon) | Nov 2025 – Dec 2025

- Led a 3-person team to architect and build an AI-powered coaching app using AWS Bedrock; planned backend scope and organized team tasks with Trello.
- Built RESTful API backend services with Flask (Python), integrating AWS Bedrock and the League of Legends API to generate personalized coaching insights.
- Resolved cross-language data formatting and API communication issues between JavaScript frontend and Python backend.
- Deployed a functional MVP to Render and demonstrated it at the hackathon.

Parallel Sorting with MPI (University Project — C) | Fall 2025

- Implemented two parallel sorting approaches in C using MPI across 6 processes with blocking and nonblocking MPI calls.
- Wrote a bash script automating 20 runs per approach and averaging MPI_Wtime results for statistically reliable benchmarks.
- Set up WSL on Windows to compile and run MPI code; applied stencil operations in a separate parallel task within the same assignment.

Cryptography Pipeline (University Project — Python) | Fall 2025

- Built an end-to-end encryption simulation in Python. Data is Huffman-encoded for compression, Hamming-encoded for error correction, then decrypted and decoded.
- Developed interactively in Jupyter Notebook; validated encoding and decoding correctness through scripted test cases.

C++ Custom 2D Game Engine (Personal Project) | Dec 2025 – Present

- Developing a 2D game engine in C++ using raylib (migrating to OpenGL) for low-level systems programming practice.
- Implemented core systems: rendering pipeline, entity-component architecture, game loop, and input management.
- Applied OOP design patterns and custom memory management; built with CMake for cross-platform compilation.

Chinese Greenhouse Simulator — Full-Stack React Web App (University Project) | Fall 2025

- Our 4-person team built the React frontend for a physics-based greenhouse simulation platform, including a simulator UI and historical data dashboard with date and location filtering.
- Headed backend development connecting frontend to MongoDB for storage and retrieval of real-time sensors and simulated data.
- Project integrates IoT sensor hardware, a Python physics simulation engine, and a React/MongoDB full-stack layer built.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C++, C, C#, SQL

Frameworks & Libraries: Flask, React, Django, Node.js, NumPy, MPI, RESTful API

Cloud & DevOps: AWS (Bedrock), Render, Git, CMake, Bash

Databases: PostgreSQL, MySQL

Operating Systems: Linux (WSL), Windows

Web: HTML, CSS